

Hotel Booking Cancellation Report

Business Problem

In recent years, City Hotel and Resort Hotel have seen high cancellation rates. Each hotel is now dealing with a number of issues as a result, including fewer revenues and less than ideal hotel room use. Consequently, lowering cancellation rates is both hotels' primary goal in order to increase their efficiency in generating revenue, and for us to offer thorough business advice to address this problem.

The analysis of hotel booking cancellations as well as other factors that have no bearing on their business and yearly revenue generation are the main topics of this report.

Assumptions

1. No unusual occurrences between 2015 and 2017 will have a substantial impact on the data used.
2. The information is still current and can be used to analyze a hotel's possible plans in an efficient manner.
3. There are no unanticipated negatives to the hotel employing any advised technique.
4. The hotels are not currently using any of the suggested solutions.
5. The biggest factor affecting the effectiveness of earning income is booking cancellations.
6. Cancellations result in vacant rooms for the booked length of time.
7. Clients make hotel reservations the same year they make cancellations.

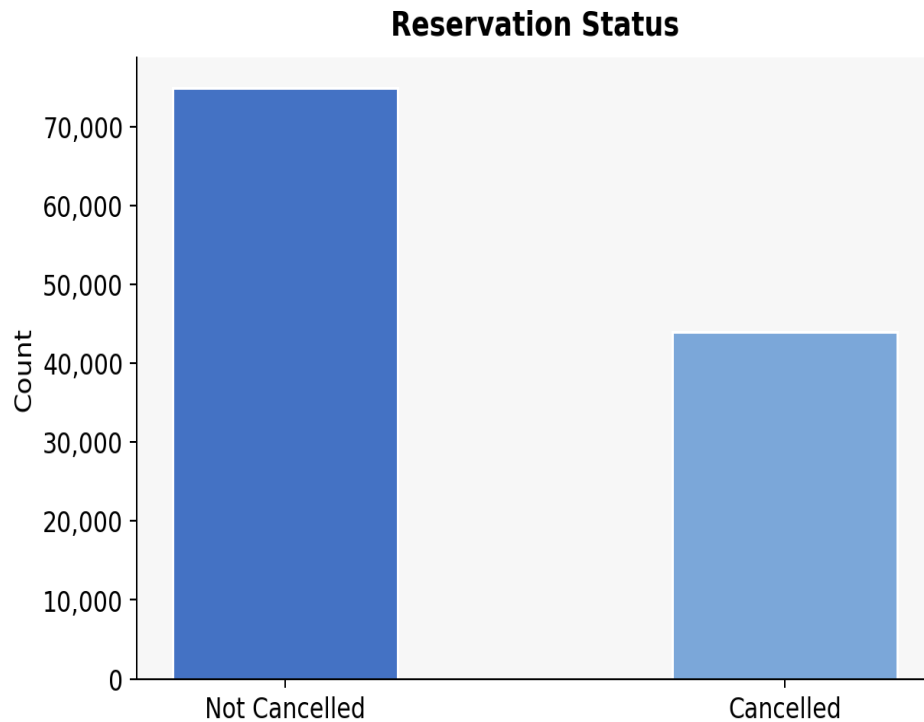
Research Question

1. What are the variables that affect hotel reservation cancellations?
2. How can we make hotel reservations cancellations better?
3. How will hotels be assisted in making pricing and promotional decisions?

Hypothesis

1. More cancellations occur when prices are higher.
2. When there is a longer waiting list, customers tend to cancel more frequently.
3. The majority of clients are coming from offline travel agents to make their reservations.

Analysis and Findings



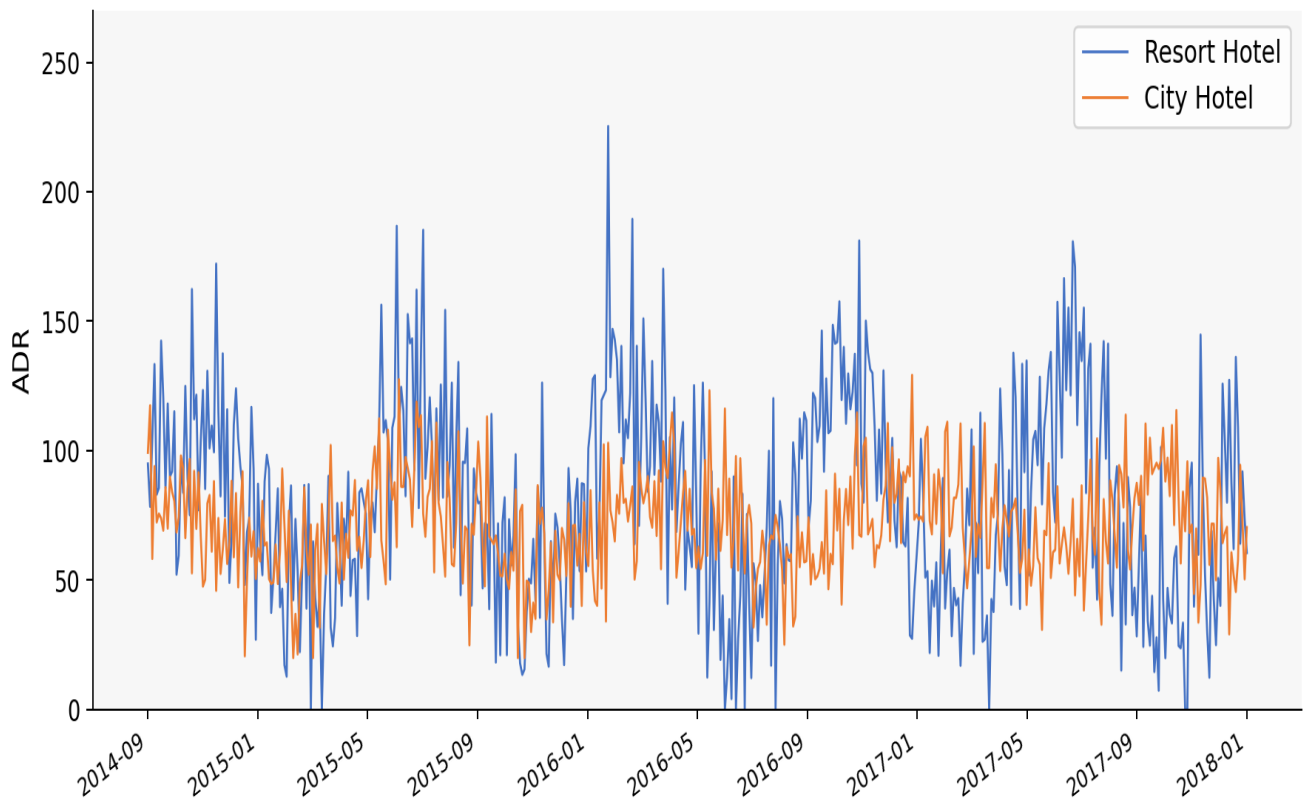
The bar graph above shows the count of reservations that were canceled versus those that were not. Despite the higher count of non-cancellations, 37% of clients still canceled their reservations — a figure that significantly impacts hotel earnings.

Reservation Status in Different Hotels

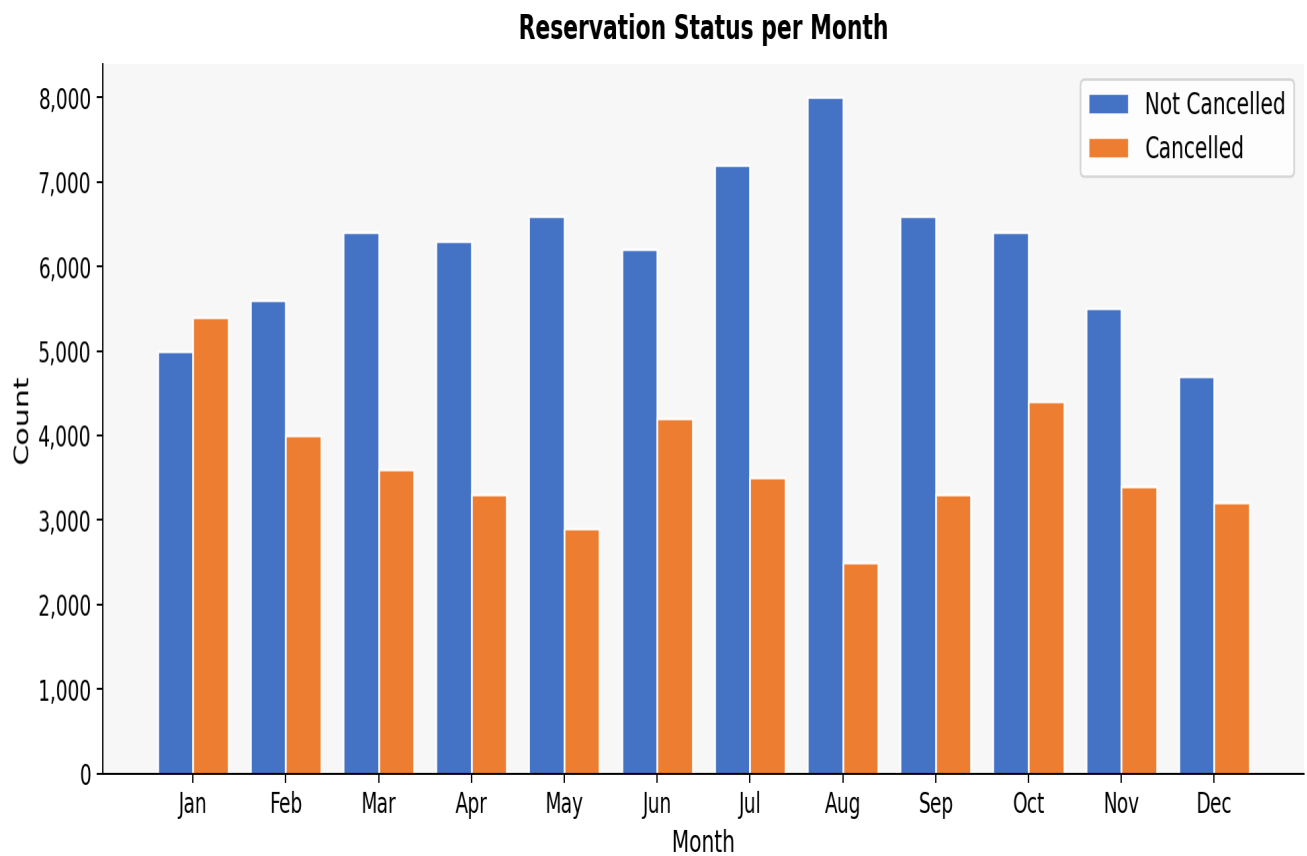


In comparison to resort hotels, city hotels have more bookings. It is possible that resort hotels are more expensive than those in cities, contributing to the higher relative cancellation proportion seen in resort properties.

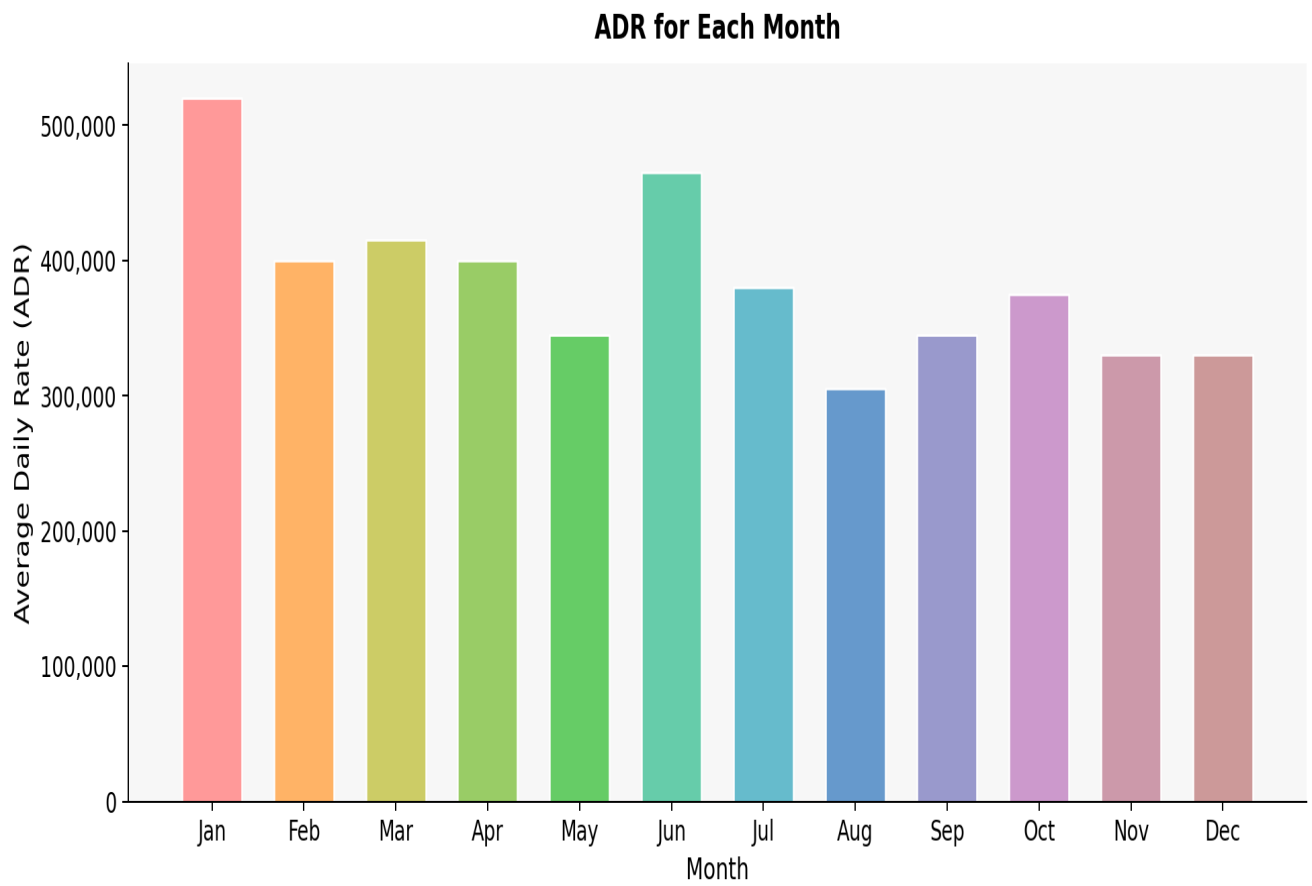
Average Daily Rate in City and Resort Hotel



The line graph above shows that, on certain days, the average daily rate for a city hotel is less than that of a resort hotel, and on other days, it is even less. It goes without saying that weekends and holidays may see a rise in resort hotel rates.



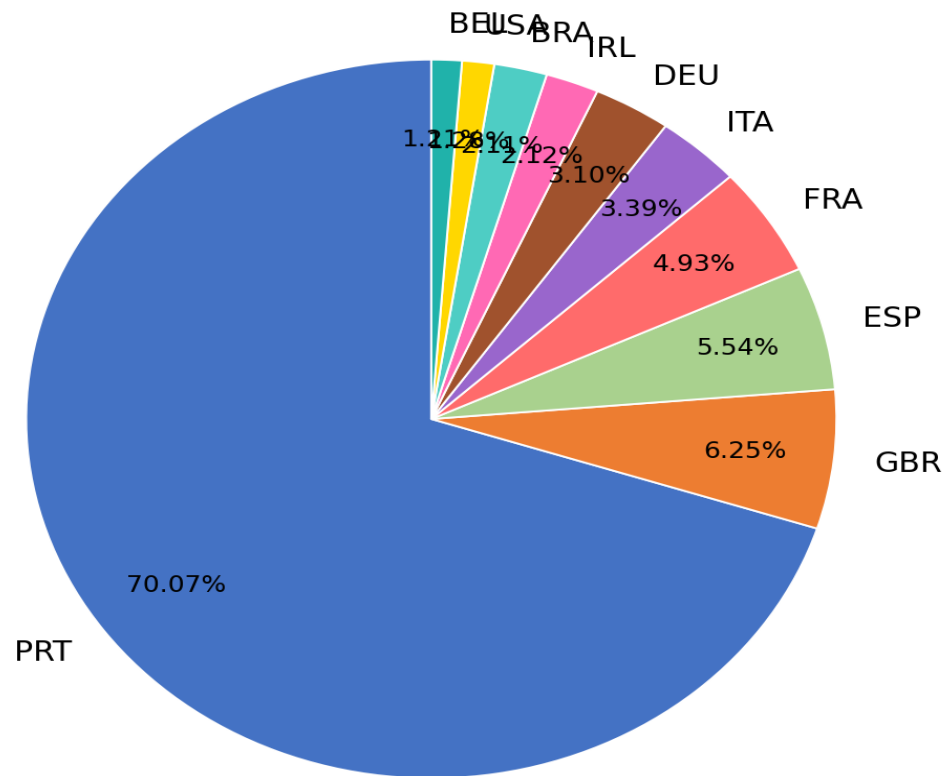
The grouped bar graph analyzes months with the highest and lowest reservation levels by reservation status. Both confirmed and canceled reservation counts peak in August. January records the highest number of canceled reservations compared to confirmations.



This bar graph demonstrates that cancellations are most common when prices are greatest and least common when they are lowest. Therefore, the cost of accommodation is a primary driver of cancellation behavior.

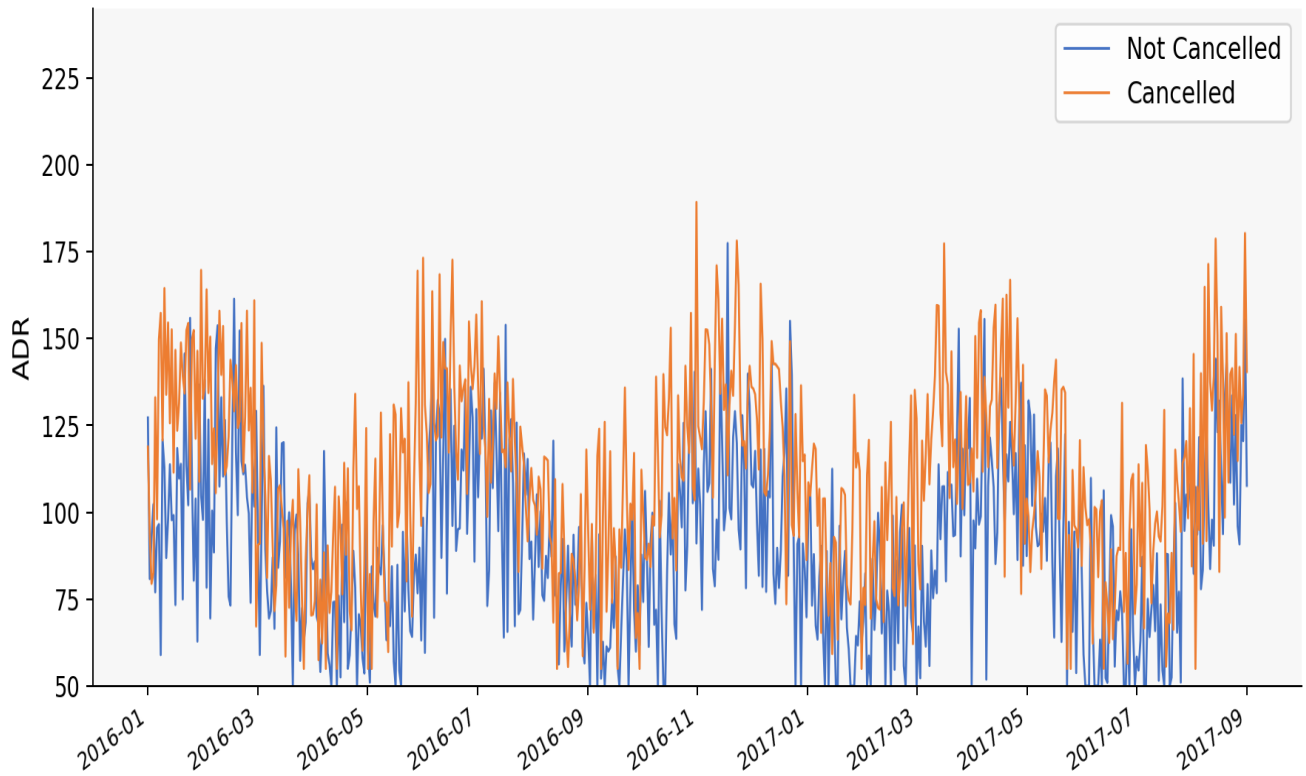
The chart below identifies the top 10 countries by number of canceled reservations. Portugal (PRT) leads by a substantial margin, accounting for over 70% of all cancellations among the top-10 countries.

Top 10 Countries with Reservation Cancelled



Regarding booking channels: around 46% of clients come from online travel agencies, 27% from groups, and only 4% book hotels directly by visiting them.

Average Daily Rate - Cancelled vs Not Cancelled



As seen in the graph, reservations are canceled when the average daily rate is higher than when they are not canceled. This confirms all the above analysis: higher prices lead to higher cancellation rates.

Suggestions

1. **Pricing Strategy:** Cancellation rates rise as the price does. In order to prevent cancellations of reservations, hotels could work on their pricing strategies and try to lower the rates for specific hotels based on locations. They can also provide some discounts to consumers.
2. **Weekend & Holiday Discounts:** As the ratio of the cancellation vs non-cancellation of resort hotels is higher than city hotels, hotels should provide a reasonable discount on room prices on weekends or on holidays.
3. **January Marketing Campaigns:** In the month of January, hotels can start campaigns or marketing with a reasonable amount to increase revenue, as cancellations are highest in this month.
4. **Quality Improvement in Portugal:** Hotels can also increase the quality of their services mainly in Portugal to reduce the cancellation rate, given Portugal's outsized share of cancellations.